

Sergei Ermolaev

✉ @J1nsei

✉ saermolaev.w@gmail.com

☎ +7 (930) 802-31-25

📍 Nizhny Novgorod (UTC+3)

🇷🇺 RU CV

Summary

Machine Learning Engineer (NLP), 3+ years in ML, including **2.5 years** of end-to-end production ML experience: RAG, LLM inference, AI Agents, multi-label classification, hybrid search. Key strengths - integrating LLMs under tight latency/resource constraints and designing pipelines from scratch through to monitoring. Open to remote work and relocation. Education - **HSE University** (AMI + Data Mining).

Experience

Proton (IT company of "AGAT" Group, *top-5 Russian car dealers* [🔗](#), 120+ showrooms)

Nizhny Novgorod

Machine Learning Engineer:

Nov 2023 – Present

- Multilabel call classification: **F1-samples 79% → 94%**, **EMR 3% → 61.6%**; sub-second CPU inference, >1000 daily calls; business impact **~\$1.7M**. Data collection and processing: transcription, cleaning, analysis, stratification in a multilabel setting; conducting experiments, production integration.
- **RAG** Assistant (RAGFlow + vLLM): **user satisfaction 4.7/5**, **TTFT p99 4.6s** under load with long context, **faithfulness 95%**. Customized framework functionality for corporate Knowledge Base specifics, stack/model selection, custom document correction and parsing pipeline, Docker, production deployment with monitoring and tracing, performance and quality benchmarking, user and support documentation.
- Ticket routing **agent** for GANDIVA (business management platform) with RAG, Guardrails, and Memory. Full development cycle, including monitoring and quality/performance evaluation (*LangGraph, RAGFlow, LangFuse, vLLM*).
- Car parts search system: reduced manual requests **300→20/day** (-93%). Database selection based on built-in capabilities and speed. Implemented **hybrid search**, custom ranking function and thresholds, wrapped in FastAPI + Docker; created abbreviation dictionary (regex + LLM + manual editing) and documentation, integration with production server.
- **LLM call labeling** (Mistral/Qwen, zero-/few-shot, **LoRA Qwen2.5-3B**): Accuracy up to 90% (LoRA), production — XLinear 88% (CPU online/offline).
- Test drive **CV tracking**: YOLOv5, multithreading, crops; accuracy 65%→87%, up to 15 fps on Consumer GPU.

Projects

HSE University Nizhny Novgorod (Academic R&D)

Sept 2022 – May

ML Engineer | *WhisperX, Llama.cpp, aiogram, guidance*. [GitHub](#) [🔗](#)

2023

- Developed an autonomous AI assistant (Telegram Bot on aiogram) for audio transcription and summarization with a focus on Data Privacy. Designed an ASR pipeline based on WhisperX with text alignment and regex noise filtering. Implemented hybrid (CPU/GPU) inference of the quantized LLM via llama.cpp with custom memory management and strict generation control using the guidance framework. The project is completely independent of external APIs and optimized for local execution on consumer hardware (from 12GB VRAM).

HSE University Nizhny Novgorod (Joint R&D project with business)

Feb 2021 – May 2021

ML Researcher | *Selenium, Colabeler, torchvision*

- Designed a data collection and preparation pipeline for a CV model detecting car damage.

Collected the dataset, performed labeling, augmentation, and class balancing.

Education

HSE University

Master's degree in Data Mining

Nizhny Novgorod, Russia

Sept 2022 – June 2024

HSE University

Bachelor's degree in Applied Mathematics and Informatics

Nizhny Novgorod, Russia

Sept 2018 – June 2022

Professional Development

[Yandex School of Data Analysis \(SHAD\) Intensive: LLM Scaling Week](#) 

Nov 10, 2025 – Nov 14, 2025

[Yandex School of Data Analysis: Agents Week](#) 

Apr 6, 2026 – Apr 10, 2026

Skills

DL/ML: Python, PyTorch, vLLM, transformers, LangChain, LangGraph, llama.cpp, deepeval, RAGFlow, NumPy, prompt engineering, infinity, unsloth, scikit-learn, WhisperX, spaCy, ultralytics

Infra/Tools: Linux, Docker, FastAPI, Git, DVC, streamlit, LangFuse, W&B, matplotlib, Grafana, Kubeflow

Data: pandas, polars, SQL

Languages: Russian - Native, English - B2 / Upper-Intermediate